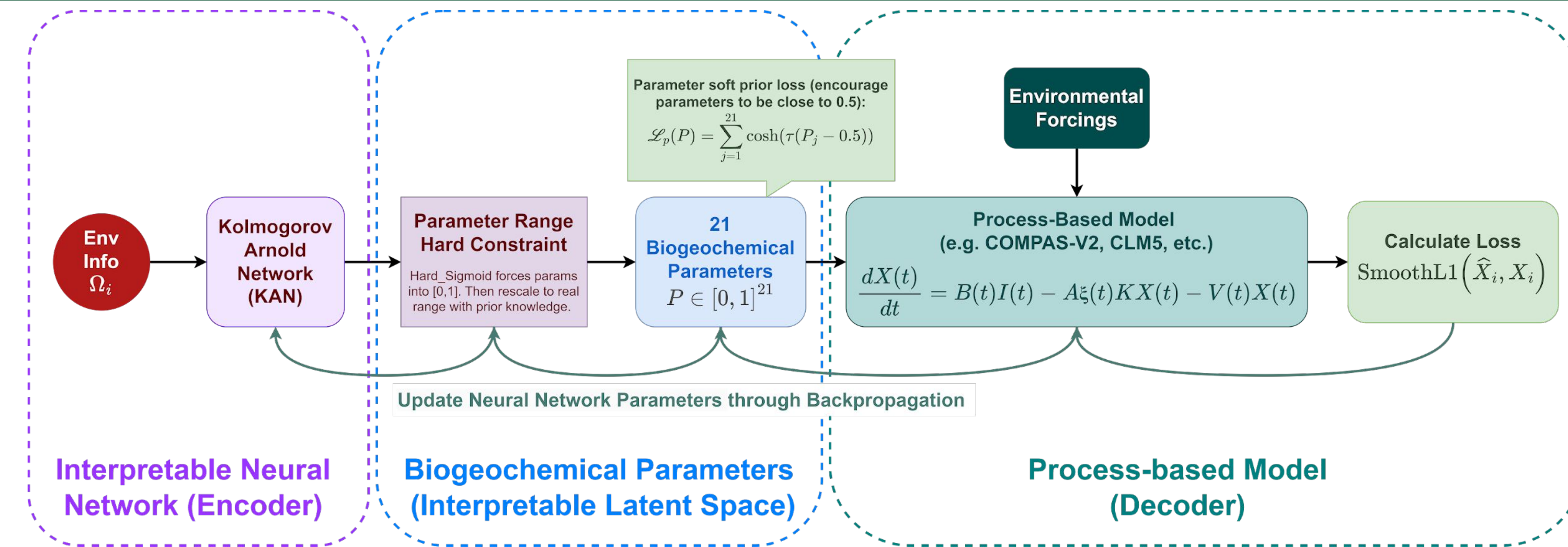
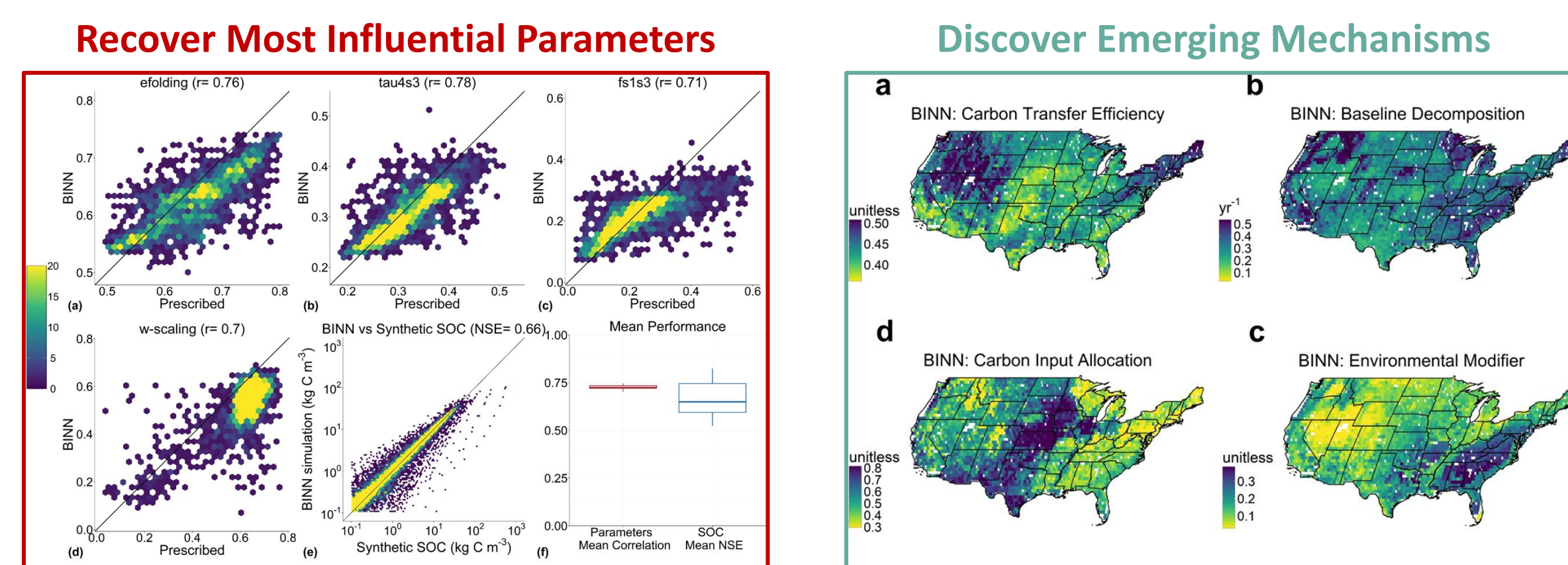


Scientifically-Interpretable Reasoning Network (SciReN) for Understanding the Soil Carbon Cycle

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Summary

- Previously we developed BINN [1], a framework to merge **scientific knowledge (process-based models)** and **data-driven learning (neural networks)** into an integrated end-to-end system
- Can predict **unobserved** biogeochemical parameters (whether in a **parameter-recovery experiment** or using **actual observations**)
- Offers superior computational efficiency and modeling accuracy compared to other data assimilation methods.
- However, BINN is not interpretable enough for scientific discovery
 - Neural network is a black-box and does not reveal functional relationships
 - Process-based model embedded (CLM5) cannot be further constrained by pool-based observations
- We make both parts more interpretable in **Scientifically-Interpretable Reasoning Network (SciReN)**, a fully-transparent model that **respects existing knowledge** and **reveals new insights**
- This framework can be applied across various fields within biogeochemistry. Here, we use it to advance modeling and understanding of soil organic carbon, since soil accounts for a large portion of the uncertainties in global carbon cycle [2]



Conclusions

- We have developed a method that
- Respects existing scientific knowledge, provided by any process-based model
 - Can learn from multiple data sources (different SOC pools, etc)
 - Reveals **new** functional relationships between environmental inputs and **unobserved biogeochemical processes**
- The entire system is trainable end-to-end, and every part of the model is fully transparent to scientists.

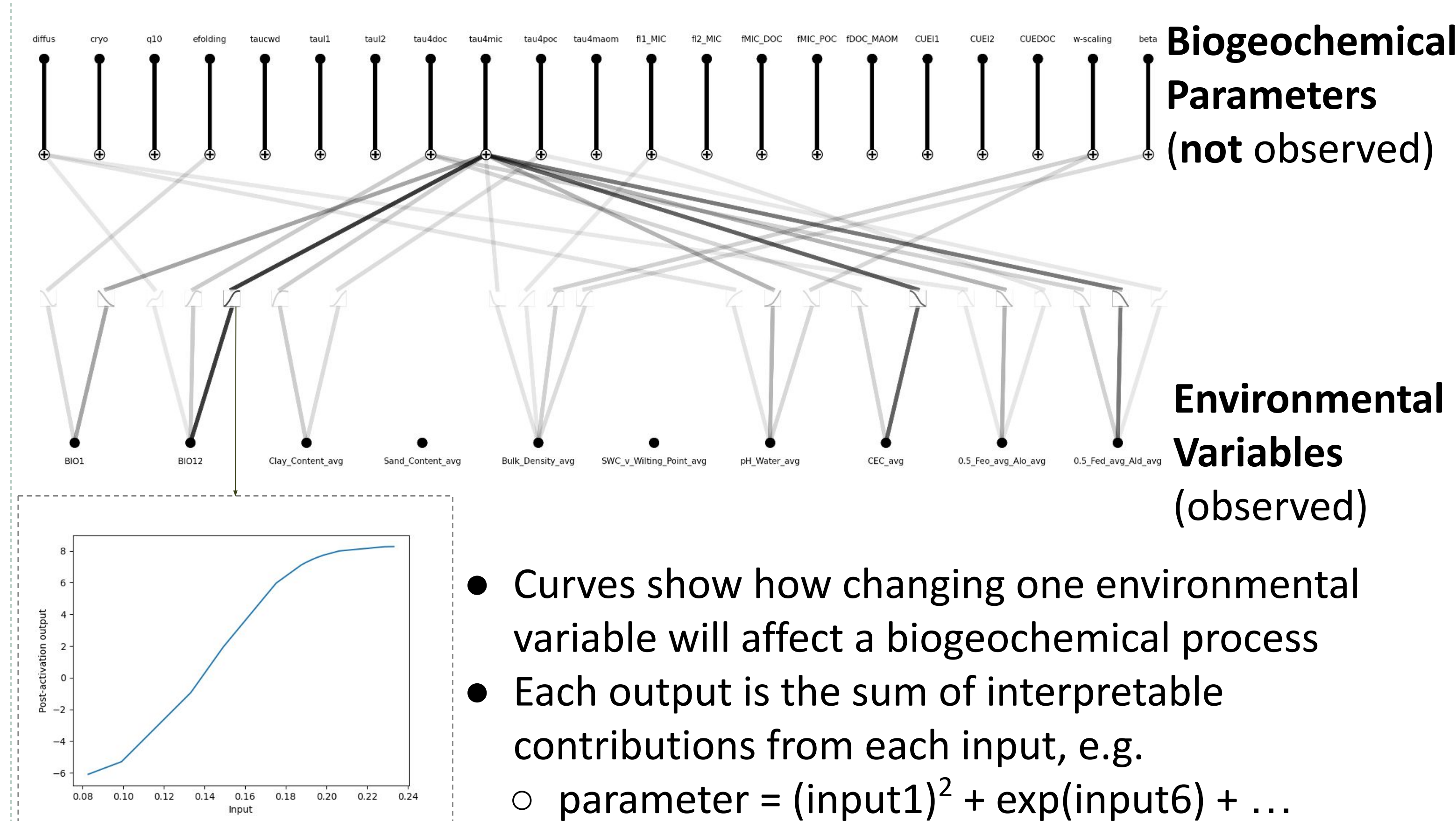
Next Steps

- Apply our techniques to various fields within biogeochemistry, such as soil inorganic carbon
- Test the functional relationships learned by the model experimentally
- Improve spatial generalization to new regions, potentially using geographic positional embeddings or domain adaptation (undergraduate research)

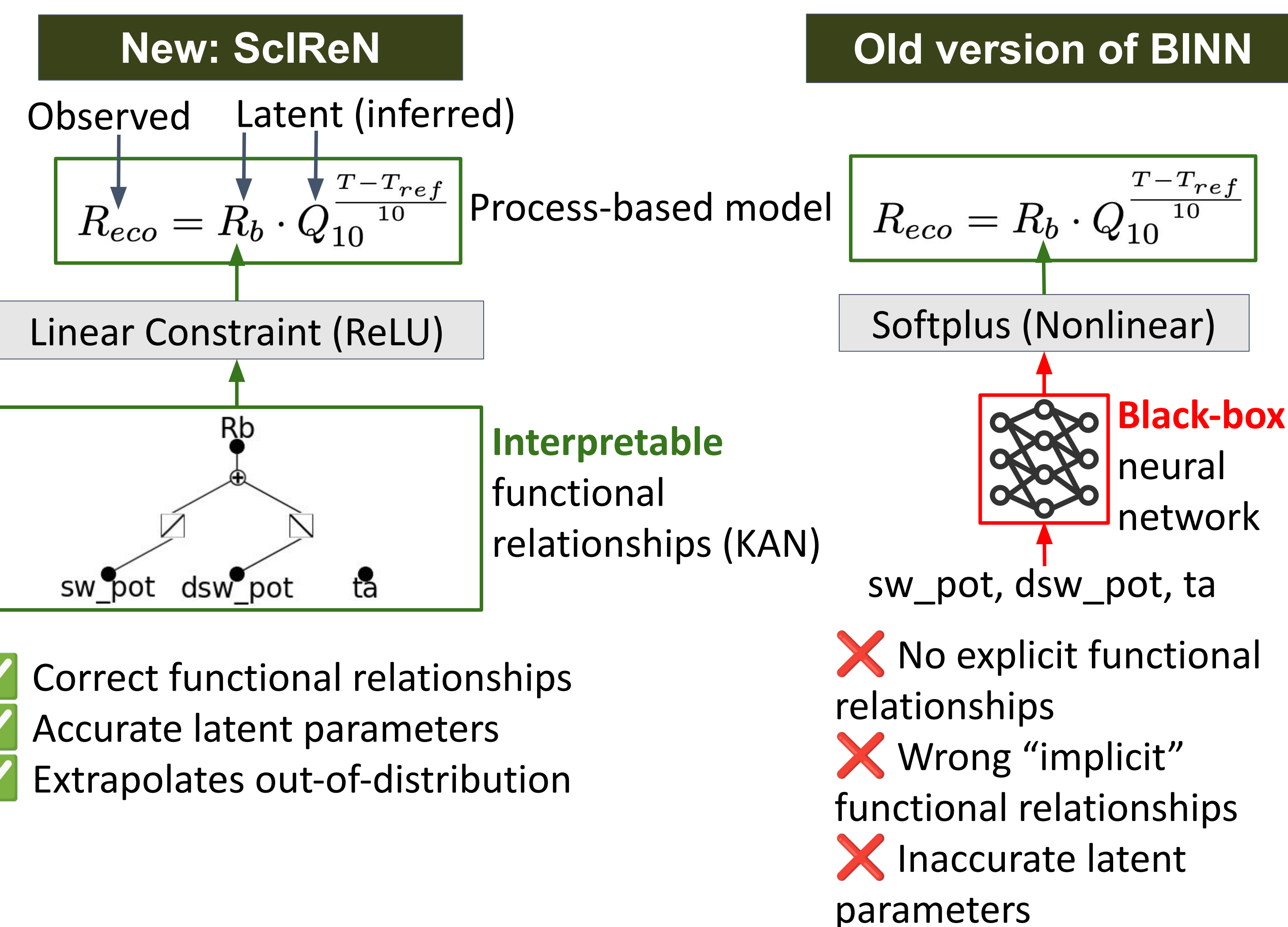
References

- [1] Xu, H., Fan, J., Tao, F., Jiang, L., You, F., Houlton, B. Z., ... & Luo, Y. (2025). Biogeochemistry-Informed Neural Network (BINN) for Improving Accuracy of Model Prediction and Scientific Understanding of Soil Organic Carbon. arXiv preprint arXiv:2502.00672. **Submitted to JGR Machine Learning and Computation.**
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- [3] Liu, Z., Wang, Y., Vaidya, S., Ruehle, F., Halverson, J., Soljačić, M., ... & Tegmark, M. (2024). Kan: Kolmogorov-arnold networks. arXiv preprint arXiv:2404.19756.

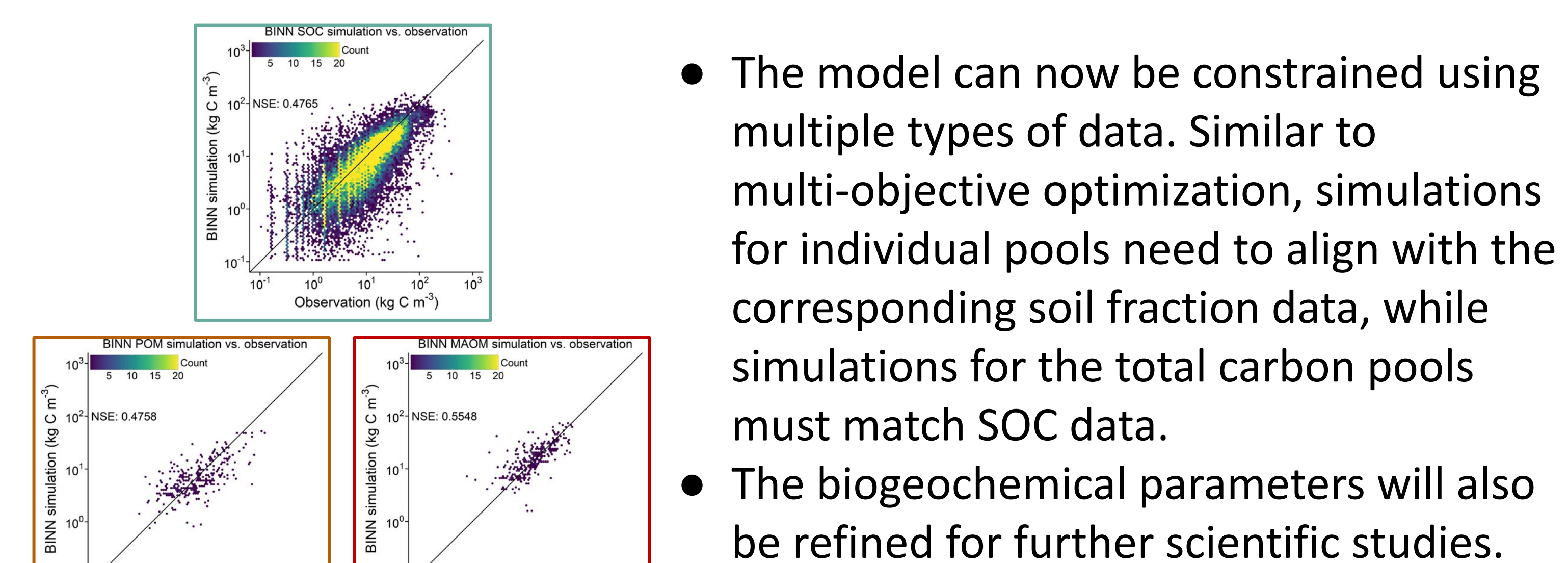
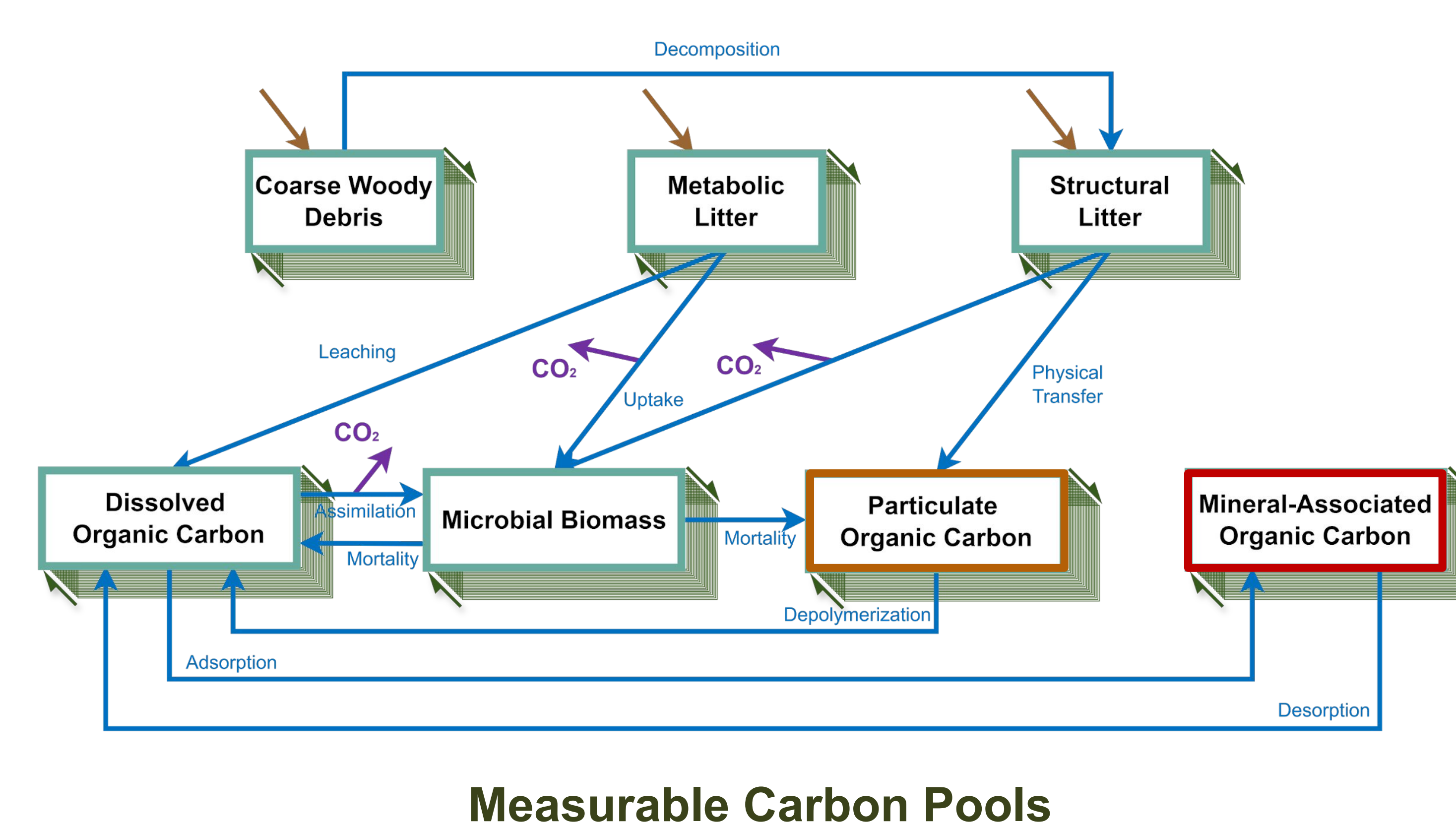
KAN Learned Interpretable Encoder (Kolmogorov-Arnold Network [3])



Applied to Ecosystem Respiration



COMPAS V2 CarbOn cycle and Microbial PARTitioning Soil model Version 2



Interpretable Biogeochemical Parameters

- A clearer understanding of each parameter, better reflecting real-world processes
- Most predicted parameters follow a normal distribution

